



Python Programming - 2301CS404

Lab - 7

Roll No : 310

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01) WAP to find sum of tuple elements.

Sample Input:

T = (10, 20, 30, 40)

Sample Output:

Sum of tuple elements = 100

```
In [10]: n = int(input("Enter the size of tuple:"))
t1 = ()
sum1=0
for i in range(n) :
    t1 += tuple(map(int,input("Enter the elements:").split()))
for i in t1:
    sum1+=i
sum1
```

Out[10]: 100

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

T = (7, 2, 9, 4, 1, 5)
K = 2

Sample Output:

Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)

```
In [14]: n = int(input("Enter the size of tuple:"))
K = int(input("Enter the value of K:"))
t1 = ()
for i in range(n) :
    t1 += tuple(map(int,input("Enter the elements:").split()))
s = tuple(sorted(t1))
min_k = tuple(s[:K])
max_k = tuple(s[-K:][::-1])
print(f"Minimum {K} elements:{min_k}")
print(f"Maximum {K} elements:{max_k}")
```

Minimum 2 elements:(1, 2)
Maximum 2 elements:(9, 7)

03) WAP to find tuples which have all elements divisible by K from a list of tuples.**Sample Input:**

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]

```
In [16]: n = int(input("Enter number of tuples: "))
tuple_list = []
for i in range(n):
    t = tuple(map(int, input("Enter elements of tuple separated by space: ").split()))
    tuple_list.append(t)
K = int(input("Enter value of K: "))
result = []
for t in tuple_list:
    if all(x % K == 0 for x in t):
        result.append(t)
print("Tuples with all elements divisible by", K, "are:", result)
```

Tuples with all elements divisible by 2 are: [(2, 4, 6), (4, 8, 12)]

04) WAP to create a list of tuples from given list having number and its cube in each tuple.**Sample Input:**

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]

```
In [18]: n = int(input("Enter number of tuples: "))
l1 = []

for i in range(n) :
    l = int(input("Enter the elements:"))
    l1.append(l)
result = [(num, num**3) for num in l1]
print(f"List of tuples with number and its cube:{result}")
```

List of tuples with number and its cube:[(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]

05) WAP to find tuples with all positive elements from the given list of tuples.**Sample Input:**

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
```

Sample Output:

Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]

```
In [19]: n = int(input("Enter number of tuples: "))
tuple_list = []
for i in range(n):
    t = tuple(map(int, input("Enter elements of tuple separated by space: ").split()))
    tuple_list.append(t)
result=[]
for t in tuple_list:
    if all(x>0 for x in t):
        result.append(t)
print(f"Tuple with all postive elements:{result}")
```

Tuple with all postive elements:[(1, 2, 3), (4, 5, 6)]

06) WAP to remove tuples of length K.**Sample Input:**

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

Sample Output:

List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8, 9)]

```
In [27]: n = int(input("Enter number of tuples: "))
tuple_list = []
for i in range(n):
    t = tuple(map(int, input("Enter elements of tuple separated by comma: ").split()))
    tuple_list.append(t)
K = int(input("Enter value of K: "))
result = []
for t in tuple_list:
    if all(len(t)!=2 for x in t):
        result.append(t)
print(f"List after removing tuples of length {K}:{result}")
```

List after removing tuples of length 2:[(3, 4, 5), (6,), (7, 8, 9)]

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

```
(5, 7, 9)
```

```
In [39]: n = int(input("Enter the size of tuple:"))
t1 = tuple()
for i in range(n) :
    t1 = tuple(map(int, input("Enter the elements: ").split()))
x = int(input("Enter the value of x:"))
T = tuple(item for item in t1 if item!=x)
print(T)
```

```
(9,)
```

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

```
(3, 5, 7, 9)
```

```
In [3]: n = int(input("Enter the size of tuple:"))
t1 = tuple()
for i in range(n) :
    t1 += tuple(map(int, input("Enter the elements: ").split()))
unique_list = []
for i in t1:
    if i not in unique_list:
        unique_list.append(i)
```

```
result = tuple(unique_list)
print(result)
```

(3, 5, 7, 9)

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

```
t = (2, 3, 4, 5)
```

Sample Output:

Resultant tuple after multiplying adjacent elements: (6, 12, 20)

```
In [4]: n = int(input("Enter the size of tuple:"))
t1 = tuple()
for i in range(n) :
    t1 += tuple(map(int, input("Enter the elements: ").split()))
result = []
for i in range(len(t1) - 1):
    result.append(t1[i] * t1[i + 1])
result_tuple = tuple(result)
print("Resultant tuple after multiplying adjacent elements:", result_tuple)
```

Resultant tuple after multiplying adjacent elements: (6, 12, 20)

10) WAP to test if the given tuple is distinct or not.

Sample Input:

```
tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
```

Sample Output:

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False
```

```
In [6]: n1 = int(input("Enter the size of tuple-1:"))
t1 = tuple()
for i in range(n1) :
    t1 += tuple(map(int, input("Enter the elements: ").split()))
n2 = int(input("Enter the size of tuple-2:"))
t2 = tuple()
for i in range(n2) :
    t2 += tuple(map(int, input("Enter the elements: ").split()))
if len(t1) == len(set(t1)):
    print(f"{t1} is distinct? -> True")
else:
    print(f"{t1} is distinct? -> False")
if len(t2) == len(set(t2)):
    print(f"{t2} is distinct? -> True")
```

```
else:
    print(f"{t2} is distinct? -> False")
```

(1, 2, 3, 4) is distinct? -> True
 (1, 2, 2, 3) is distinct? -> False

11) WAP to rotate the elements of a tuple to the right by **k** positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

```
In [7]: n = int(input("Enter the size of tuple-1:"))
t1 = tuple()
for i in range(n) :
    t1 += tuple(map(int, input("Enter the elements: ").split()))
k = int(input("Enter value of k: "))

k = k % len(t1)
rotated = t1[-k:] + t1[:-k]
print(rotated)
```

```
(40, 50, 10, 20, 30)
```

12) WAP to merge two tuples of equal length alternately.

Sample Input:

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

```
In [8]: n1 = int(input("Enter the size of tuple-1:"))
t1 = tuple()
for i in range(n1) :
    t1 += tuple(map(int, input("Enter the elements: ").split()))
n2 = int(input("Enter the size of tuple-2:"))
t2 = tuple()
for i in range(n2) :
    t2 += tuple(map(int, input("Enter the elements: ").split()))
result = []
for i in range(len(t1)) :
    result.append(t1[i])
    result.append(t2[i])
print(tuple(result))
```

(1, 2, 3, 4, 5, 6)

13) WAP to create a tuple of squares of elements that are even and greater than 10.

Sample Input:

T = (4, 12, 7, 18, 10)

Sample Output:

(144, 324)

```
In [9]: n = int(input("Enter the size of tuple-1:"))
t1 = tuple()
for i in range(n) :
    t1 += tuple(map(int, input("Enter the elements: ").split()))
result = []
for i in t1:
    if i % 2 == 0 and i > 10:
        result.append(i * i)
print(tuple(result))
```

(144, 324)

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

T = (4, 7, 9, 11, 15)

Sample Output:

((1, 7), (3, 11))

```
In [10]: n = int(input("Enter the size of tuple-1:"))
t1 = tuple()
for i in range(n) :
    t1 += tuple(map(int, input("Enter the elements: ").split()))
result = []
for index, value in enumerate(t1):
    if value > 1:
        is_prime = True
        for i in range(2, int(value ** 0.5) + 1):
            if value % i == 0:
                is_prime = False
                break
        if is_prime:
            result.append((index, value))
print(tuple(result))
```

((1, 7), (3, 11))

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

```
T = ('a', 'b', 'c', 'd', 'e')
```

Sample Input:

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

```
In [15]: n = int(input("Enter the size of tuple:"))
t1 = tuple()
for i in range(n) :
    t1 += tuple(map(str,input("Enter the elements: ").split()))
result_even = []
result_odd = []
for i in range(len(t1)):
    if i % 2 == 0 :
        result_even.append(t1[i])
    else :
        result_odd.append(t1[i])
print(f"even = {tuple(result_even)}")
print(f"odd = {tuple(result_odd)}")
```

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

16) WAP to compute column-wise sum from a tuple of lists.**Sample Input:**

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
```

Sample Output:

```
12 15 18
```

```
In [16]: n = int(input("Enter number of lists: "))
t = []
for i in range(n):
    lst = list(map(int, input("Enter list elements: ").split()))
    t.append(lst)
t = tuple(t)
cols = len(t[0])
result = []
for j in range(cols):
    col_sum = 0
    for i in range(len(t)):
        col_sum += t[i][j]
    result.append(col_sum)
for val in result:
    print(val, end=" ")
```

```
12 15 18
```